

Linear Probing Provides Robust and Efficient Detection of Machine-Generated Text

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Abstract

Distinguishing machine-generated text (MGT) from human-written text (HWT) becomes increasingly important due to potential misuse. However, most supervised detectors often degrade out-of-domain (OOD) and require large, diverse training sets. In this work, we analyze the linearity and quality of MGT representations and show that simple linear probes outperform a wide range of detectors while being substantially more sample-efficient. We first show that MGT and HWT latent representations are linearly separable in low-dimensional space, and provide a plausible explanation for this separability through systematic differences in their representation quality. Motivated by these insights, we train two variants of simple linear probes and evaluate them across 4 benchmarks against 16 baselines. Probes consistently improve OOD detection (+11 AUC), requiring solely <100 samples to reach near-peak performance. We show that this transferability arises because probes recover a shared latent MGT direction that generalizes across diverse settings. Finally, we demonstrate that probing vectors capture a continuous spectrum of “machineness”, highlighting their potential for fine-grained estimation of AI-edited text. Overall, our work provides insights into latent-space differences between MGT and HWT and demonstrates the potential of linear probes as as robust and sample-efficient MGT detectors. We release our code on [github](#).

1 Introduction

Large language models (LLMs) now generate human-like text at scale, and their output has spread rapidly across platforms such as social media (Sun et al., 2025) or academic writing (Liang et al., 2024). This proliferation raises the risk of potential misuse, including misinformation campaigns (Goldstein et al., 2023) or plagiarism (Hutson, 2024). In response, a growing line

of work builds detectors that separate machine-generated text (MGT) from human-written text (HWT) (Crothers et al., 2023; Wu et al., 2025a).

Prior work groups MGT detectors into two families (Wu et al., 2025a): zero-shot methods (e.g., Hans et al., 2024; Bao et al., 2023) and supervised classifiers (e.g., Hu et al., 2023; Verma et al., 2024).¹ While supervised approaches dominate recent benchmarks (Macko et al., 2025; Quaremba et al., 2026), their practical utility remains limited: they often generalize poorly to out-of-domain (OOD) settings (Wu et al., 2025a; Doughman et al., 2025) and require large, diverse training corpora (Hu et al., 2023). Additionally, a recent line of work leverages how LLMs rewrite machine- versus human-written text differently (Mao et al., 2024; Wu et al., 2025b), but adds substantial overhead from generating rewrites.

A promising direction for overcoming these limitations lies in the Linear Representation Hypothesis (LRH) (Park et al., 2023), which posits that high-level concepts are encoded as linear directions in latent space. Recent work has identified such directions for concepts including truth (Marks and Tegmark, 2024) or political ideology (Kim et al., 2025). This raises a natural question: *Are machine- and human-written text linearly separable in latent space, and if so, why? How can we exploit this structure for enhanced MGT detection?*

In this work, we empirically study the latent geometry of HWT and MGT and show that simple linear probes are sufficient to improve the robustness and sample efficiency of MGT detection. We begin by visualizing that HWT and MGT are largely linearly separable in low-dimensional space, which we corroborate by *reduced* detection performance when increasing the complexity of nonlinear probes (§3.1). We then investigate *why* this linearity emerges by analyzing both populations with

¹We focus on passive detection and exclude watermarking.

four representation quality metrics (Skean et al., 2025): compared to HWT, MGT representations collapse into a compressed, more anisotropic, and lower-dimensional subspace (§3.2). These systematic differences provide a plausible explanation for why a simple hyperplane can effectively separate the two in latent space.

Leveraging these insights, we train **MGT probes**: simple linear probes (Alain and Bengio, 2016; Nanda et al., 2023) on frozen LM representations to detect MGT. We consider two variants: layer-wise probes and a single probe trained on representations concatenated across layers. Crucially, both variants operate on PCA-reduced features, retaining only 2.4% of the original dimensions with minimal performance loss. The resulting probing vector defines a linear “machineness” direction in latent space, which we use for detection by projecting test activations onto it (Tigges et al., 2024). Across 4 benchmarks spanning 16 evaluation settings and compared against 16 baseline detectors, MGT probes improve in-domain detection by up to 18 AUC (§5.2) and 11 AUC OOD (§5.3).

To explain this strong generalization, we analyze probing vectors learned across diverse datasets and settings. We find that the vectors are highly aligned, indicating that probes recover a shared latent MGT direction (§5.4). Moreover, probes achieve strong performance with only 10–100 training samples and exhibit substantially lower sampling uncertainty than training-based baselines (§5.5). Together, our findings suggest that MGT is encoded as a compact, shared, and linearly accessible direction in latent space. This allows simple linear probes to generalize better while requiring fewer examples than more complex supervised detectors.

Lastly, we examine whether probing vectors can serve as a continuous measure of AI editing in text (Zhang et al., 2024) (§5.6). We find strong correlations between probe projection scores and edit-strength metrics (Saha and Feizi, 2025), meaning that more extensively AI-edited text is projected closer to the MGT subspace. This finding suggests that the latent MGT direction encodes a *continuous* spectrum of “machineness,” which can be leveraged for fine-grained estimation of AI-edited text.

Our contributions are threefold:

1. We find that human- and machine-written text are linearly separable in latent space and explain this separability through systematic differences in their representation geometry.

2. We introduce two simple linear probe variants that recover this latent direction, improving OOD generalization over supervised detectors (+11 AUC) while exhibiting greater reliability and requiring only 10–100 training examples.
3. We show that the learned MGT direction encodes a *continuous* spectrum of “machineness,” enabling fine-grained estimation of AI editing strength through probe projection scores.

2 Related Work

Linear Representation Hypothesis (LRH) Recent work has provided growing support for the LRH, which posits that high-level concepts are represented as linear directions in latent space (Park et al., 2023). Various studies have successfully used linear probing to identify concepts such as spatial reasoning (Nanda et al., 2023), truth (Marks and Tegmark, 2023), sentiment (Tigges et al., 2024), or political ideology (Kim et al., 2025). To identify probing vectors, one typically construct a contrastive dataset containing positive and negative instances of the concept of interest (e.g., MGT and HWT). Common approaches for identifying probing directions include difference-in-means (Belrose, 2023) and linear probes (Alain and Bengio, 2016). The resulting probing vector represents the concept as a linear direction in activation space and can be used for detection, interpretation, or representation steering (Zou et al., 2025).

MGT Detectors MGT detectors can be broadly categorized as zero-shot or supervised methods (Wu et al., 2024). Zero-shot detectors are training-free and typically exploit discriminative differences in token probabilities (Bao et al., 2023; Su et al., 2023; Hans et al., 2024) or leverage the observation that LLMs rewrite MGT and HWT differently (Wu et al., 2025b; Zhu et al., 2023a; Yang et al., 2024). Most supervised detectors formulate MGT detection as a binary classification task and train machine-learning classifiers (Hu et al., 2023; Guo et al., 2024; Mao et al., 2024; Thai et al., 2026). Compared to prior latent-representation-based detectors, our approach operates directly on activations rather than a single intrinsic-dimensionality feature (Tulchinskii et al., 2023), uses *linear* probes across layers rather than a single-layer *nonlinear* classifier (Yu et al., 2024), and learns probes in a *low-dimensional* space rather than identifying

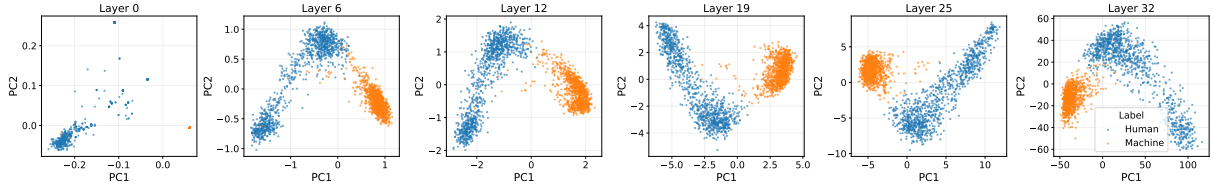


Figure 1: Projections of Wikipedia human- and machine-text hidden states onto the first two principal components across layers. **Takeaway:** Machine- and human-text representations become linearly separable as early as layer 6, and this separation remains consistent through the final layers. Appendix A confirms the same pattern across other domains.

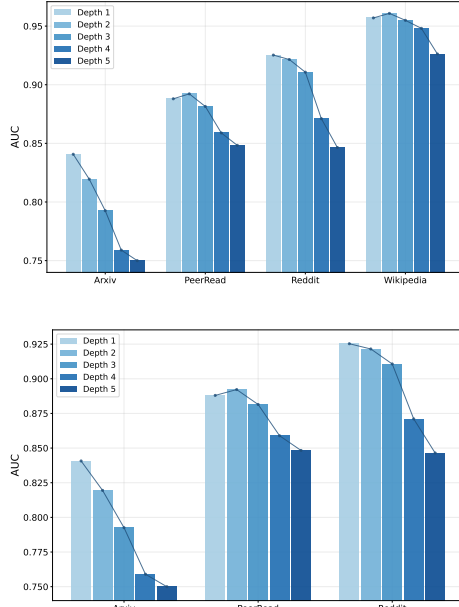


Figure 2: In-domain (top) and OOD (bottom) AUC scores for probing middle-layer representations with MLPs of increasing levels of non-linearity. **Takeaway:** Increasing model complexity consistently reduces performance, supporting the hypothesis that machine- and human-text representations are *linearly* separable in latent space.

directions in a *training-free* manner in the *full* representation space (Chen et al., 2025). We review current MGT detectors and provide a detailed comparison with related work in Appendix B.

3 On the Separability and Quality of MGT Representations

In this section, we first examine the *linear separability* of HWT and MGT representations (§3.1) and then analyze their *representation quality* to characterize their latent-space differences (§3.2). We conduct all analyses on the Wikipedia subset of M4GT (Wang et al., 2024) using Llama-8B (Grattafiori et al., 2024) as the backbone model. Appendix A.2 shows that our findings gen-

eralize across domains.

3.1 Linear Separability

We begin by asking: *Are HWT and MGT linearly separable in latent space?* While prior work has shown that linear classifiers can work (Tulchinskii et al., 2023; Chen et al., 2025),² it remains unclear whether the underlying decision boundary is truly linear or whether more complex nonlinear boundaries are required.

MGT and HWT are linearly separable in low-dimensional latent spaces. Figure 1 visualizes residual stream activations of MGT and HWT across layers, projected onto the first two principal components. While the two distributions overlap in the first layer, they become clearly *linearly* separable as early as layer 6. This separation remains remarkably stable throughout the remaining layers. Appendix Figure 8 further shows that each domain and generator forms distinct local subspaces, within which HWT and MGT remain linearly separable.

To test whether the decision boundary is indeed linear, we train MLP probes of increasing complexity on mid-layer representations to predict MGT (Gurnee and Tegmark, 2024). If nonlinear structure were important for detection, deeper classifiers should improve performance. Instead, Figure 2 shows that detection performance deteriorates as nonlinearity increases in both ID and OOD settings. Our findings empirically support that MGT and HWT are separated by a predominantly linear decision boundary in latent space.

3.2 Representation Quality

Yet, it remains unclear which representational differences make this linear separation possible. We therefore ask: *How do HWT and MGT differ in latent space?* To this end, we analyze their latent

²We have included the detailed comparison with them in Appendix B.3.